AI-Powered Unified Sri Lankan Tourism Platform Using GraphRAG and LLM Orchestration

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

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ABSTRACT

This research develops a route risk analysis and vehicle recommendation component for tourism journeys in Sri Lanka. The aim is to connect available road and historical hazard evidence with traveler constraints and explain the resulting recommendations. A dataset of 598 records from 22 route groups supports a ten-feature severity classifier. Four candidate models are compared using grouped development cross-validation, followed by repeated validation of shortlisted models. Gradient Boosting is selected and achieves a recorded final holdout Macro F1 of 0.8780 and accuracy of 0.8820. These values measure agreement with severity-derived dataset labels, not accident probability. At runtime, the component integrates route retrieval, historical context, classifier inference, and deterministic vehicle filtering by budget, passenger capacity, category, and known gradient suitability. The supplied demonstration uses Kandy to Colombo, four passengers, a LKR 25,000 budget, and the Sedan category. It displays a 112.98-kilometre route, a High severity result, and a Hyundai Verna (Turbo) estimate of LKR 22,701. Alternative lower-risk comparison is unavailable in this example. The individual contribution is the integration of suitable route and vehicle selection with recommendation explanations. The principal limitations are undocumented source-data provenance, incomplete road-profile coverage, approximate route-family matching, and the absence of independent field or stakeholder validation. The findings support a working advisory prototype and an internal classification experiment, while broader claims require verified data and further evaluation.

Keywords: route risk, vehicle recommendation, tourism, Gradient Boosting, explainability

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
CSV	Comma-separated values
CV	Cross-validation
F1	Harmonic mean of precision and recall
LKR	Sri Lankan rupee
ML	Machine learning
OSRM	Open Source Routing Machine
POI	Point of interest

1 INTRODUCTION

1.1 Overall project context

The R26-IT-050 project develops a tourism application for Sri Lanka with several connected planning functions. The Route & Vehicle Advisor supports the transport decision within that application. It combines the requested journey with available road and historical hazard evidence and with the traveler's budget, passenger count, and preferred vehicle category. Destination selection, itinerary optimisation, and accommodation functions provide wider project context but are outside the individual contribution examined here.

1.2 Individual component context

This thesis covers the Route & Vehicle Advisor associated with IT22555458. It examines route-level historical severity classification, evidence-supported comparison, vehicle suitability, and explanations. Destination relevance, itinerary generation, and accommodation recommendation are adjacent project functions. The individual focus is selecting a suitable route and vehicle and explaining why they fit the journey.

1.3 Problem statement

How can a tourism planning system combine historical route hazard evidence, available road characteristics, and traveler constraints to provide an explainable route classification and vehicle recommendation while making incomplete data visible?

The problem includes both prediction and decision logic. A classifier assigns a severity category from available features. Separate rules determine whether a vehicle satisfies the traveler's constraints. The classification does not establish that a recommended vehicle prevents accidents, and a feasible vehicle recommendation does not establish that the route is safe.

1.4 Research gap

The gap addressed by this component is the absence, within the project's basic journey-selection workflow, of one traceable decision process that connects route

evidence with vehicle constraints. Providing a route and a list of vehicles independently leaves the traveler to interpret whether the available vehicle attributes are appropriate for the road profile.

The implementation addresses this integration gap by retaining the evidence behind a route classification and the rules behind a vehicle recommendation. It also represents incomplete evidence explicitly. This distinction matters because an unmeasured gradient cannot justify the same suitability conclusion as a known gradient. The contribution is an implemented research prototype; a claim of superiority over commercial travel platforms would require a defined benchmark and comparative study.

1.5 Aim and objectives

To develop and evaluate an explainable route risk analysis and vehicle recommendation component for tourism journeys in Sri Lanka using the project's available historical hazard, road-profile, and vehicle data.

• Prepare a structured dataset while preserving missing road measurements and excluding direct target leakage.

• Compare candidate classifiers using route-group separation and reserve a final holdout for evaluation after model selection.

• Integrate route retrieval, historical graph evidence, classifier inference, and current weather context.

• Filter and rank vehicles using budget, passenger capacity, category, and known gradient suitability.

• Expose recommendation reasons, data limitations, and controlled dependency failures.

• Define further evaluation needed to assess field performance and user understanding.

1.6 Research questions or hypotheses

RQ1: How effectively do candidate classifiers reproduce the available historical severity labels when training and evaluation use separate route groups?

RQ2: How can the component combine route evidence and traveler constraints to select a feasible vehicle and expose its selection rules?

RQ3: What limitations become visible when the prototype presents route, vehicle, and historical-evidence outputs in a journey demonstration? The study treats these as research questions; it does not assert statistically tested hypotheses about accident reduction.

1.7 Scope limitations and assumptions

The scope includes route-level historical severity classification, evidence-supported alternative comparison, vehicle filtering and ranking, and explanations. It assumes that the supplied vehicle attributes and baseline prices are usable research inputs, while their provenance and currency remain to be verified. Road-family aggregates approximate the requested route; exact traversed-segment analysis is not available. Current weather is contextual. The system does not forecast accidents, verify future road conditions, or guarantee journey safety.

1.8 Individual contribution

The individual contribution of WIJESINGHE I.D.W. (IT22555458) concerns suitable route and vehicle selection and explanations of those recommendations. Its technical scope includes feature preparation, the risk-classification workflow, route evidence integration, constraint-based vehicle selection, and presentation of reasons and limitations. Shared integration boundaries should be reconciled with the team's agreed allocation before final submission.

1.9 Thesis structure

Chapters 2 and 3 cover related work and research methodology. Chapters 4 and 5 describe design and implementation. Chapter 6 presents results and original screenshots. Chapters 7 and 8 discuss findings and future work. Appendices provide reproducibility notes and disclosure of report-preparation assistance.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This is a focused narrative review of methods relevant to the implemented component, rather than a systematic review claiming exhaustive coverage. It considers primary work on gradient boosting and road-network routing, together with official documentation for the libraries used in the project [1], [2], [3], [4], [5], [6]. Sources were selected for their relationship to prediction, grouped evaluation, and route retrieval. No study-selection counts or formal quality-scoring protocol were recorded. A broader empirical review of tourism vehicle recommendation is still required to support a field-wide novelty claim.

2.2 Domain and theoretical foundations

Route retrieval, risk classification, and vehicle suitability are separate technical tasks. A routing engine calculates paths through a road network. A supervised classifier learns associations between feature values and labels. A recommendation rule filters or orders candidates according to explicit constraints. Combining these tasks does not make their outputs interchangeable: route duration is not hazard probability, and vehicle capacity is not evidence of crash protection.

Gradient boosting builds an additive predictive model through successive fitting steps [5]. For this study, the target is a historical severity category. The scikit-learn implementation supplies the classifier used in the reproducible training pipeline [2].

2.3 Related research

Friedman presents gradient boosting as a framework for function approximation using a stagewise additive model [5]. This provides a methodological foundation for the selected classifier, but the paper does not validate the project's Sri Lankan labels or its vehicle recommendations. The project therefore evaluates its candidate classifiers on its own grouped dataset.

Geisberger and colleagues describe contraction hierarchies for efficient routing in road networks [6]. This work concerns the computational routing problem. It does not establish that a shortest or fastest route is safer. In the present component, the

routing provider supplies candidates, after which a separate severity model and evidence-matching process support advisory comparison.

2.4 Existing systems products and techniques

OSRM exposes route distance, duration, geometry, and optional route alternatives [1]. These outputs support the journey display, while the project's own rules determine whether distinct historical evidence is available for comparison. Alternative retrieval can produce only one usable route.

Scikit-learn GroupKFold separates groups between folds and GroupShuffleSplit creates randomized splits over groups [3], [4]. They support evaluation on held-out route families. Neo4j is used in the implementation to retrieve historical context, followed by deterministic explanations [7]. No performance comparison with commercial routing or rental platforms was conducted.

2.5 Gap synthesis and conceptual framework

The practical gap is the need to connect retrieved route evidence with budget, capacity, category, and gradient suitability in one explainable workflow. The conceptual sequence is journey request, route retrieval, evidence matching, severity classification, vehicle constraint checking, ranking, and explanation. Evidence availability accompanies the result. This integration is the proposed component contribution; the available review does not establish that no other system combines comparable functions.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a prototype-development approach with offline classification experiments and software-level verification. The process comprises dataset preparation, model comparison, backend integration, vehicle selection, and presentation of results. The stored evaluation artifacts support analysis of the classifier. Application source and regression tests provide evidence of implemented behavior. Neither evidence source establishes effectiveness in real travel conditions.

The offline pipeline prepares the training table, fits preprocessing within model pipelines, compares candidate architectures, evaluates the selected architecture on a reserved holdout, and refits it on the complete dataset for deployment. The online workflow accepts a journey request, retrieves route context, constructs features, calls the deployed classifier, and applies vehicle constraints [7], [8].

3.2 Data participants or study materials

The repository contains primary road, disaster, and vehicle CSV files, processed datasets, a training dataset, and a cleaning report. These are existing project inputs; the available evidence does not establish that the author personally collected the original observations. Source organizations, collection dates, licenses, sampling procedures, and the observed or generated nature of these records are not documented sufficiently to make stronger claims [7].

The road records contain attributes such as gradient, elevation, friction, terrain, surface, and width. Disaster records supply route-associated hazard context and source severity labels. Vehicle records provide attributes used for capacity, gradient suitability, ranking, and hire estimates. These sources serve different functions and should retain separate provenance records.

A defensible final data description requires the source of each dataset, the date obtained, permitted uses, and a clear account of any simulation, manual editing, or augmentation. Until that information is available, model performance must be interpreted as performance on the supplied research dataset.

The prepared dataset contains 598 records from 22 route groups. The cleaning report records 113 exact matches, 22 route-level matches, and 463 records without matching road data. Thus, only 135 records, or 22.6 percent, have matching road profiles. The remaining 77.4 percent constrain how confidently road-specific behavior can be interpreted [9].

Dataset characteristic	Recorded value
Training records	598
Route groups	22
High severity records	317
Medium severity records	224
Low severity records	57
Exact road matches	113
Route-level road matches	22
Records without road matches	463

Table 3.1. Dataset composition from the stored cleaning report and component documentation.

The class distribution is uneven. Low severity has substantially fewer records than High or Medium severity. For this reason, overall accuracy should be considered alongside Macro F1, which gives each class equal weight when averaging class-specific F1 scores. Per-class behavior and the reliability of the labels remain important even when the aggregate score is high.

No human-participant study is reported. The supplied application screenshots constitute a demonstration of visible software behavior, not an independent user study.

3.3 Experimental or study procedure

The dataset-building script normalizes textual values, extracts route codes and segment information, converts numeric values, and repairs identified source-row formatting issues. The stored cleaning report records 27 repaired rows. The target risk_level is derived from the source Severity Level field, mapping valid values into

Low, Medium, and High. The original severity field must therefore remain outside the model inputs to avoid direct target leakage [8].

Road matching first uses available segment evidence and then route-level evidence. Where no corresponding road profile is available, numeric measurements remain missing. Unknown categorical characteristics are represented explicitly. Imputation is performed later within the modeling pipeline; an imputed model input is not a newly observed road measurement.

The five numeric inputs are gradient, elevation, friction, historical_occurrence_count, and road_data_available. The five categorical inputs are terrain, road_surface, road_width, hazard_type, and season. The route_code field defines evaluation groups and is excluded from predictive inputs [8].

Numeric preprocessing uses median imputation followed by standardization. Categorical preprocessing uses a most-frequent imputer and one-hot encoding with unknown categories ignored. Each candidate uses the same preprocessing structure within a pipeline. Fitting preprocessing within training partitions helps avoid using validation information when estimating transformations.

The availability flag informs the classifier that road data may be missing, but it does not recover absent measurements. A future ablation experiment should compare the complete model with models that omit road attributes or the availability indicator. That comparison would help identify how much of the recorded performance depends on hazard context, missingness patterns, or measured road characteristics. No such ablation result is claimed here.

Four classifiers are compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. They offer different predictive structures while sharing the same feature schema. The training script fixes random_state at 42 for the main split and model configurations [8].

One GroupShuffleSplit holdout is created before model comparison using a test-size parameter of 0.20. This parameter refers to the proportion of groups, so the held-out fraction of records need not equal exactly 20 percent [4]. The holdout is excluded from cross-validation and repeated development validation.

The remaining development routes are used in five-fold GroupKFold cross-validation. The two models with the strongest mean Macro F1 are shortlisted. Those candidates then undergo ten repeated GroupShuffleSplit validations using development data only. The selection rule prioritizes mean validation Macro F1, with mean train-validation gap as a tie-breaker. The selected architecture is evaluated once on the reserved holdout, then refit on all 598 records to create the deployment artifact [8].

3.4 Measures and evaluation metrics

Accuracy is the proportion of correct predicted labels. For a given class, precision measures how many predictions for that class are correct, and recall measures how many records belonging to that class are identified. F1 is the harmonic mean of precision and recall. Macro F1 averages the three class-specific F1 scores equally.

The report distinguishes cross-validation, repeated development validation, and final holdout results. Standard deviations describe variation across the recorded development splits. They should not be interpreted as confidence intervals for nationwide deployment. A single final holdout also does not establish how performance varies across other independently sampled regions or time periods.

3.5 Data analysis

Candidate means and standard deviations are compared on development folds. The top two candidates proceed to repeated development validation, and selection follows mean validation Macro F1 with mean train-validation gap as a tie-breaker [8]. Final holdout results are interpreted separately from model selection. No statistical-significance test is reported for differences between models.

The screenshot demonstration is assessed descriptively: entered constraints are compared with the displayed route, vehicle capacity, category, and estimated cost. A displayed suitability label is treated as an application output, not a separately verified engineering assessment. The evidence distinguishes implementation behavior from externally validated outcomes.

3.6 Ethics privacy security and responsible AI

The intended benefit is better-informed journey planning. The principal communication risk is that users may interpret a low-severity class as a guarantee. Explanations must therefore expose incomplete evidence and the meaning of confidence. Location requests may be transmitted to external geocoding and routing providers. Production use requires appropriate notices, retention limits, access control, and credential handling. These are deployment requirements; this report does not claim a completed privacy or security audit.

The demonstration uses city-level locations and a passenger count. No identifiable participant or driver study is presented. The candidate must verify source-data permissions and applicable institutional requirements before submission. Appendix B records the report-drafting assistance used in this chat; it is not a declaration of all tools used throughout the research project.

3.7 Reproducibility and research artefacts

The repository preserves dataset-construction and training scripts, the deployed model artifact, evaluation CSV files, application code, and regression tests [7], [8], [9], [10], [11]. The recorded training workflow uses random_state 42 and a fixed feature schema. Versioned dependencies and a compatible Python environment are needed to repeat inference. This report reproduces stored experiment outputs; it does not claim a new training run. The original PNG screenshots are embedded without cropping, retouching, annotation, or recompression. They are scaled proportionally only for document placement.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The functional requirements are to validate a journey request, retrieve route context, classify available inputs, retrieve historical evidence when possible, recommend a feasible vehicle, and display the reasoning and limitations. Route alternatives should be compared only when the available evidence supports a meaningful comparison.

Non-functional requirements include understandable outputs, reproducible evaluation, bounded calls, and controlled failures. Successful CSV parses are cached within a backend process. Runtime performance still requires measurement.

Condition	Expected behavior
Missing or invalid required inputs	Reject the request with validation feedback
Unavailable Python inference	Legacy analysis returns ML UNAVAILABLE; alternative-route analysis may retain the route with risk unavailable
Unavailable Neo4j	Continue with graph status marked unavailable
Unavailable weather	Preserve unavailable context rather than imply no rain
Unknown road gradient	Preserve uncertainty and suppress the capability upsell
No feasible vehicle	Display an explicit absence of a recommendation

Table 4.1. Selected implementation requirements and failure behavior.

4.2 Component architecture within the complete solution

The frontend captures origin, destination, budget, passenger count, and preferred vehicle category. The Node.js backend validates these inputs and coordinates routing, available historical evidence, Python inference, and deterministic vehicle selection. The Python predictor is launched as a bounded subprocess for a request. It loads the model within that process, so process startup and artifact loading remain part of runtime overhead [7].

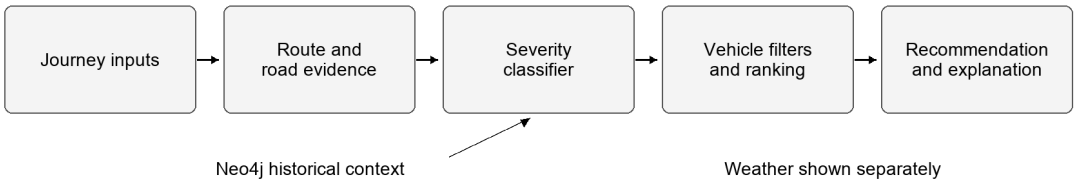


Figure 4.1. Main data flow of the Route & Vehicle Advisor. Weather is contextual; Neo4j evidence can populate named classifier inputs.

Location handling uses a verification chain involving Nominatim, Geoapify candidates, and optional Ollama candidate generation. Suggested location corrections are checked before acceptance. This workflow can reject uncertain matches and does not guarantee that every misspelling or ambiguous place name can be resolved [7].

The current road CSV has no route geometry or latitude and longitude fields. Consequently, the implementation aggregates a matched road family rather than measuring the exact traversed segments. It uses maximum known gradient, average numeric elevation and friction, and deterministic modes for categorical road attributes. The result includes the aggregation type and segment count.

4.3 Design alternatives and decisions

The principal design decision is to keep prediction and recommendation separate. Machine learning assigns a severity class, while explicit rules enforce vehicle feasibility. This makes the budget and capacity decisions inspectable. A single opaque score combining weather, price, road features, and historical data would be harder to interpret and is not the deployed design.

A second decision is to prefer route-group evaluation over an unrestricted record-level split. This reduces direct overlap of road families across fitting and evaluation. A third decision is to expose incomplete route comparison instead of generating unsupported rankings. The component can retain an analyzed route while explaining that alternative comparison is unavailable.

4.4 Detailed design

OSRM candidates are normalized into distance, duration, geometry, and road-label evidence. Candidates are matched conservatively using explicit road codes or sufficient meaningful name tokens. Where at least two distinct candidates have

evaluable evidence, the implementation compares risk class, High-class probability, Medium-class probability, duration, distance, and route identifier in that order. Candidates sharing the same evidence family cannot justify claims of distinct risk evidence [7].

With fewer than two evaluable candidates, the response uses the default-analyzed-route state. A single evaluable candidate may be selected; if none is evaluable, the implementation attempts analysis of the default route. This state does not establish a comparative risk advantage. Neo4j supplies historical context and can populate occurrence count, hazard type, and season. Its current reasoning is deterministic and template-based after retrieval. Current weather is returned separately and does not alter the classifier through an arbitrary multiplier.

Vehicle recommendation is a deterministic decision process informed by the route analysis. Candidate vehicles must satisfy the requested passenger capacity, category, and budget. When road gradient is known, vehicle gradeability is compared with the required gradient. The difference is represented as a gradeability margin [7].

For Low risk, the implemented comparison prioritizes estimated hire price and then gradeability margin. For High risk, it prioritizes gradeability margin, torque, engine capacity, and then price. For the intermediate case, the comparison prioritizes margin, torque, and price. These rules are transparent heuristics based on the available dataset attributes; they are not experimentally established crash-safety rankings.

If gradient is unknown, the system does not treat it as an observed zero-gradient road. Unknown gradient does not reject vehicles and suppresses the higher-capability upsell. This behavior avoids presenting unsupported suitability evidence while allowing a constrained recommendation to continue.

The selected route's analysis is passed internally to the vehicle helper. Route safety values are not accepted back from the client as authoritative model results. If no vehicle satisfies the constraints, the interface must represent the absence of a recommendation rather than inventing a feasible vehicle.

The model-specific estimate is calculated as BaseHireCharge plus DistanceKM multiplied by RentalPricePerKM. The API exposes these values and the total in a structured pricing object [7].

For illustration only, a hypothetical base charge of LKR 3,000 and a rate of LKR 100 per kilometre produce LKR 13,000 for a 100-kilometre journey. These numbers demonstrate the formula and are not a recorded vehicle quote or a current market rate.

The repository does not establish verified providers and effective dates for the baseline rates. Therefore, the report describes them as dataset estimates. An operational service would require sourced prices, effective dates, and rules for charges that are outside the current formula.

The response separates model inputs, contextual weather, graph retrieval status, vehicle constraints, and ranking rules. This makes the recommendation traceable: a reviewer can inspect what was supplied to the model and why a vehicle passed a rule. The trace is not a causal explanation of accidents and does not establish a proven safety outcome.

4.5 Standards and professional practice

The component uses structured API responses and validates required inputs before analysis. The same feature names are maintained between training and inference. Risk values used by vehicle selection are passed internally rather than accepted as authoritative client-supplied results. Source attribution is retained on the map, including the visible OpenStreetMap contributor credit. Dependency failure states and data limitations are part of the response contract [7], [8].

The report distinguishes class confidence, dataset accuracy, and real-world risk. Original screenshots remain intact so their visible messages can be examined against the written interpretation. These practices support traceability; no formal standards certification or completed security audit is claimed.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The frontend uses React Native and Expo, including a web presentation. The backend uses Node.js and Express, with Python for classifier inference. The machine-learning pipeline uses pandas, joblib, and scikit-learn; the documented model compatibility version is scikit-learn 1.7.2 [7]. OSRM provides route candidates, geocoding services resolve locations, and Neo4j provides historical context when available. Current weather is a separate enrichment service.

5.2 Component implementation

The recorded workflow selects Gradient Boosting. Its configured parameters include 150 estimators, a learning rate of 0.05, and maximum tree depth of 3. The deployment path uses risk_model_v2.joblib, loaded by predict_safety.py. The prediction response includes the predicted class, class probabilities, model metadata, and the input feature values used for that request [7], [8].

Confidence is the model's probability assigned to its predicted class. It is not the model's holdout accuracy and is not a calibrated probability that a traveler will experience an accident. The current workflow does not include a probability-calibration experiment. User-facing descriptions should retain this distinction.

The result interface presents the analyzed route, map access, trip context, vehicle candidates, historical explanation, and technical details. The screen sequence in Section 6.2 documents these outputs for the supplied Kandy-to-Colombo example.

5.3 Integration with the other components

The component accepts journey information from the broader application and can consume an ordered itinerary produced elsewhere. Shared backend helpers support consistent vehicle rules across individual-route and itinerary workflows. This integration does not transfer ownership of destination selection or itinerary optimisation to this component. The demonstration in Chapter 6 concerns a single origin-destination request and does not independently demonstrate the complete multi-component itinerary flow.

5.4 Verification and software testing

The repository includes backend regression tests for request validation, dependency failures, explanation traces, unavailable gradient behavior, and itinerary integration. These tests provide repeatable checks of the application contract. The technical notes record an earlier passing test checkpoint; the complete suite was not rerun during preparation of this report. A final submission should attach the command, environment, date, and complete output from a fresh execution [7], [11].

Software tests check application behavior; machine-learning evaluation checks predictive labels. Neither replaces the other, and neither establishes field effectiveness.

The current screenshots document a successful interface path. They do not show invalid-input handling or a no-matching-vehicle scenario. Those cases should be supported by dated automated test output or additional test evidence before making a comprehensive validation claim.

5.5 Deployment and operation

For local operation, the backend is started with npm start from the backend directory, and the browser frontend with npm run web from the frontend directory. The wider application's separate AI service is started with npm run ai-service. Environment settings must point the frontend to reachable services. Optional graph and weather dependencies require their own configured availability.

The supplied screenshots show the web application at localhost:8081 on 11 October 2026. This establishes the displayed local demonstration context, not a production deployment or an uptime result.

6 RESULTS AND VALIDATION

6.1 Research results

Table 6.1 reproduces the development cross-validation results stored in model_comparison_results.csv. These are existing experiment outputs, not results from a new training run performed for this report [10].

Candidate classifier	CV Macro F1	Standard deviation	CV accuracy
Gradient Boosting	0.7742	0.0682	0.7916
Random Forest	0.7723	0.0649	0.7895
Logistic Regression	0.7286	0.0379	0.7460
Decision Tree	0.7176	0.0725	0.7471

Table 6.1. Five-fold grouped cross-validation on development data.

Gradient Boosting and Random Forest have similar cross-validation means and entered the repeated-validation shortlist. Their small difference in cross-validation Macro F1 does not establish statistically significant superiority. The recorded selection rule uses the subsequent development stability results.

Shortlisted model	Mean validation Macro F1	Standard deviation	Mean training gap
Gradient Boosting	0.8408	0.0332	0.1111
Random Forest	0.8218	0.0357	0.0822

Table 6.2. Ten repeated grouped validations using development data only.

Gradient Boosting has the higher repeated validation mean, while Random Forest has the smaller mean train-validation gap. The implemented rule selects Gradient Boosting based on mean validation Macro F1. This decision reflects the predefined criterion; it does not imply that Gradient Boosting dominates every measure.

The selected model's final holdout Macro F1 is 0.8780 and accuracy is 0.8820. Macro precision is 0.8562 and macro recall is 0.9074 [10]. Only the selected architecture has final holdout results in the comparison file. The final artifact is subsequently refit on the complete dataset, so these metrics describe the reserved-

holdout experiment rather than a second independent evaluation of the refitted artifact.

The holdout score exceeds the development cross-validation mean. Different route compositions can produce different difficulty levels, so this should not be described as a guaranteed performance improvement. The recorded data support a successful internal classification experiment, with external validation still outstanding.

6.2 Product and technical results

Figures 6.1-6.7 present the original screenshots supplied by the author. The input example specifies a budget of LKR 25,000, four passengers, Kandy as the origin, Colombo as the destination, and Sedan as the category. The screenshots demonstrate displayed application behavior. They do not establish real-trip outcomes or validate the underlying historical records.

6.2.1 Journey input

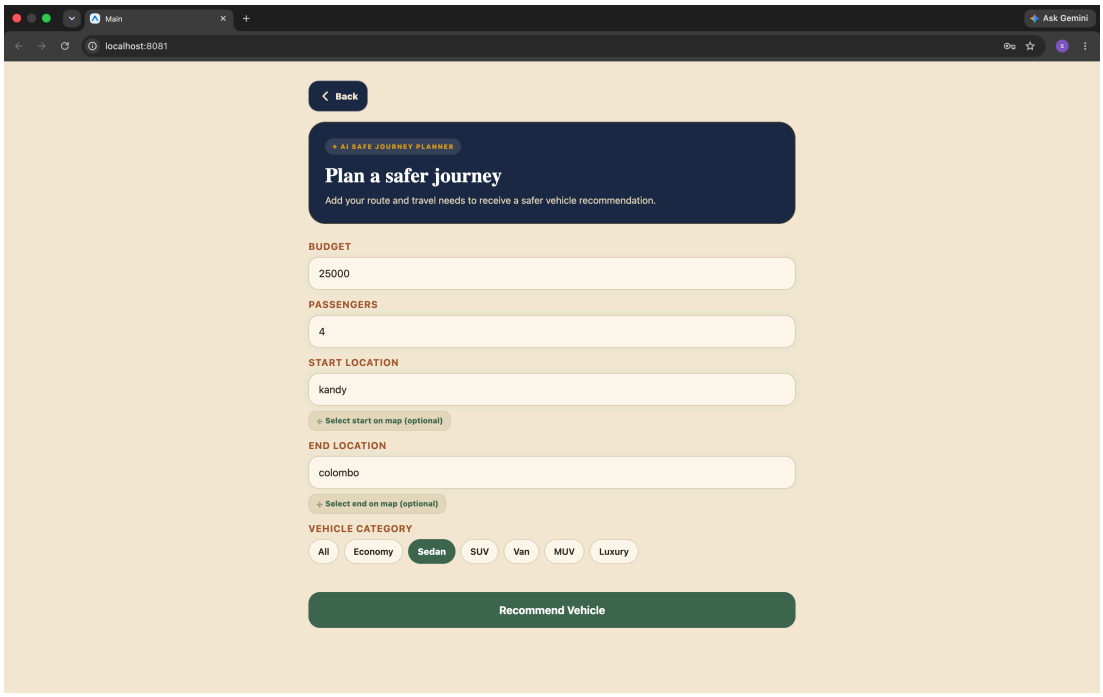


Figure 6.1. Completed input form for the Kandy to Colombo demonstration.

Figure 6.1 records the supplied inputs: LKR 25,000, four passengers, Kandy to Colombo, and Sedan. These values establish the constraints against which the visible recommendations can be interpreted.

6.2.2 Route map

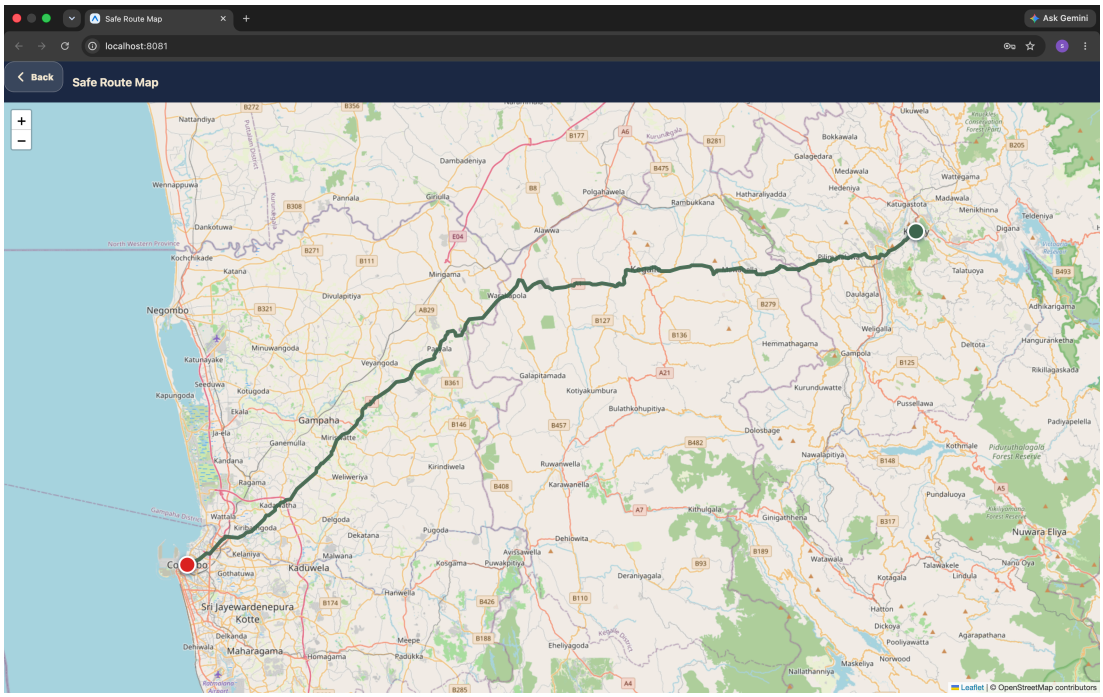


Figure 6.2. Displayed route geometry between Kandy and Colombo.

Figure 6.2 shows the original map view and its OpenStreetMap attribution. The map documents the displayed route geometry. It does not establish a successful comparison among alternative routes.

6.2.3 Route analysis and trip context

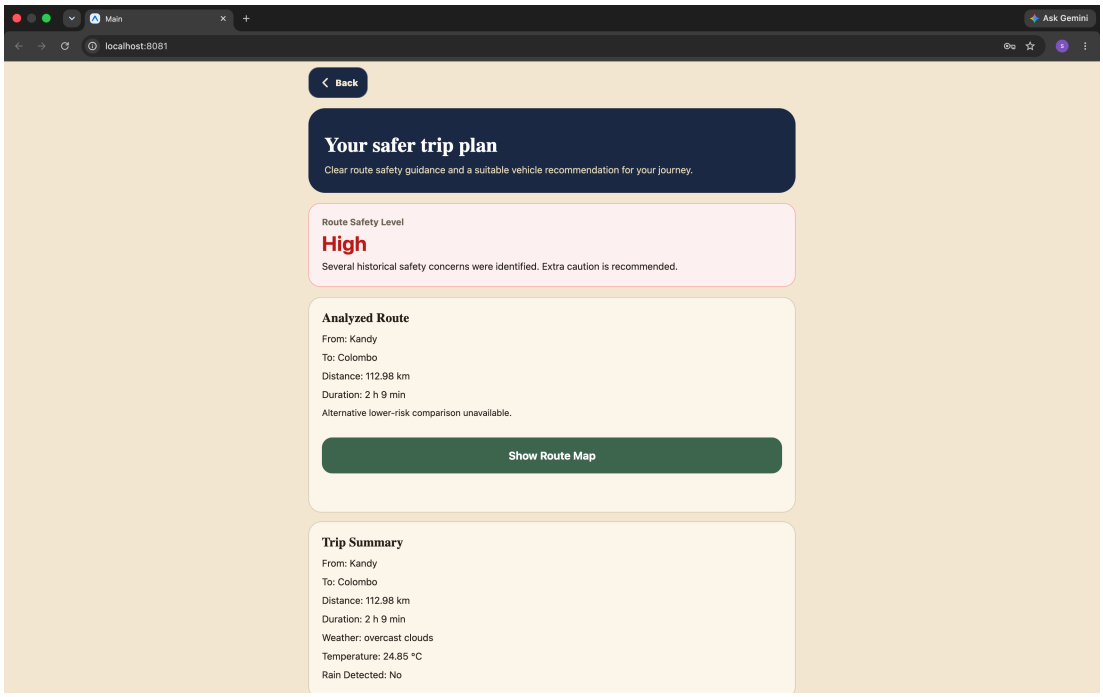


Figure 6.3. High severity output with route distance duration and current weather.

Figure 6.3 displays Kandy to Colombo, 112.98 km, and an estimated duration of 2 h 9 min. The result states that alternative lower-risk comparison is unavailable. High is interpreted as the historical severity class, not a claim of high safety. The weather panel is current context, not a forecast for a future departure.

6.2.4 Vehicle shortlist

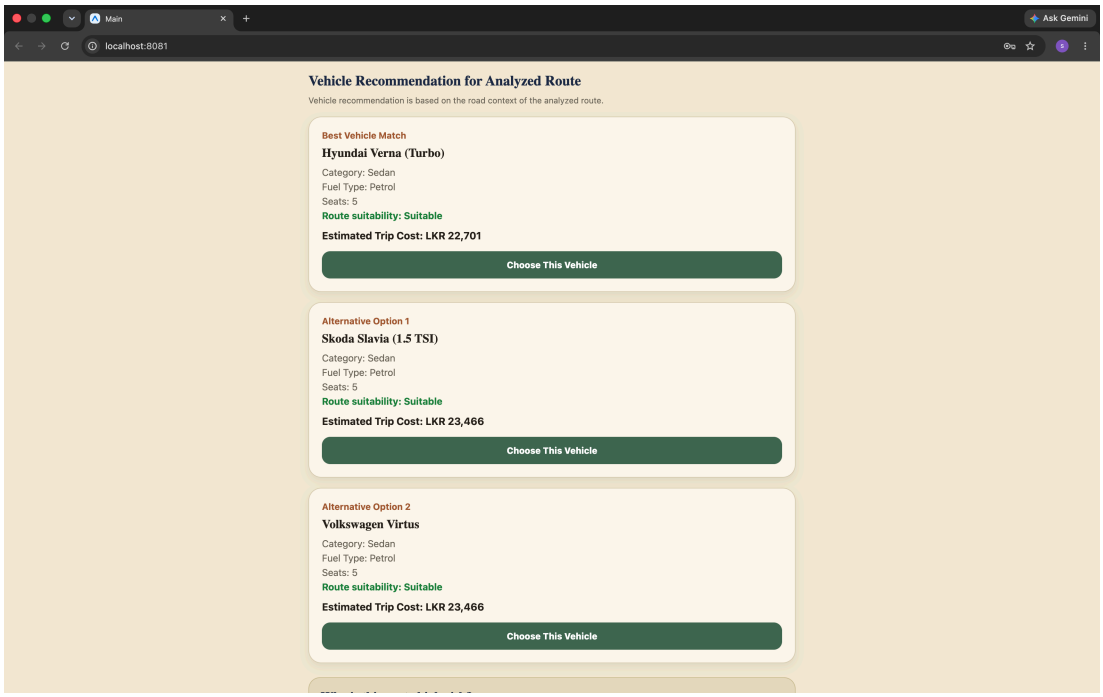


Figure 6.4. Recommended sedan and two alternative vehicle options.

Figure 6.4 shows Hyundai Verna (Turbo) as the best match at LKR 22,701, with five seats. Its displayed category matches Sedan, its seats meet the four-passenger request, and its estimate is LKR 2,299 below the stated budget. Skoda Slavia (1.5 TSI) and Volkswagen Virtus are shown at LKR 23,466. The cards display Suitable, but do not expose the complete gradeability and ranking calculations.

6.2.5 Historical explanation

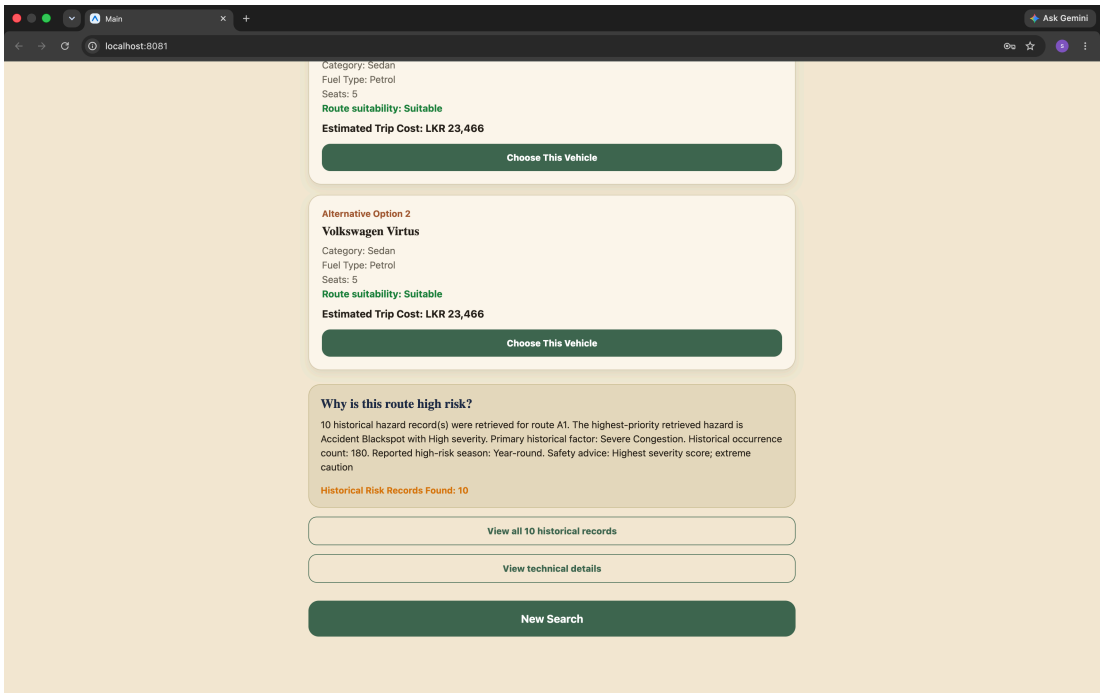


Figure 6.5. Historical context displayed beneath the vehicle recommendations.

Figure 6.5 shows the interface explanation for the High result, including 10 retrieved historical records for route A1 and the highest-priority retrieved hazard. The displayed count of 180 historical occurrences is distinct from the 10 retrieved records. These are application-retrieved dataset values, not newly verified field observations.

6.2.6 Retrieved historical records

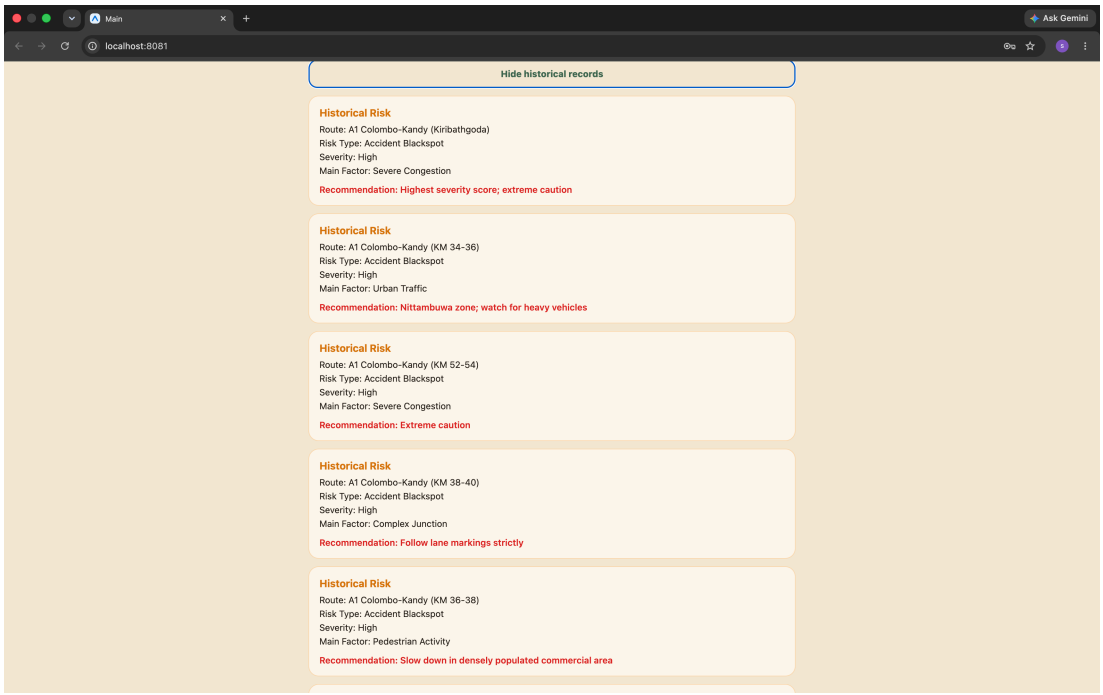


Figure 6.6. Expanded historical risk records and advisory text.

Figure 6.6 shows several retrieved A1 records, their hazard types, severity labels, contributing factors, and advisory text. The screenshot provides evidence of retrieval and presentation. It does not independently verify the records or prove they match each exact traversed road segment.

6.2.7 Technical prediction details

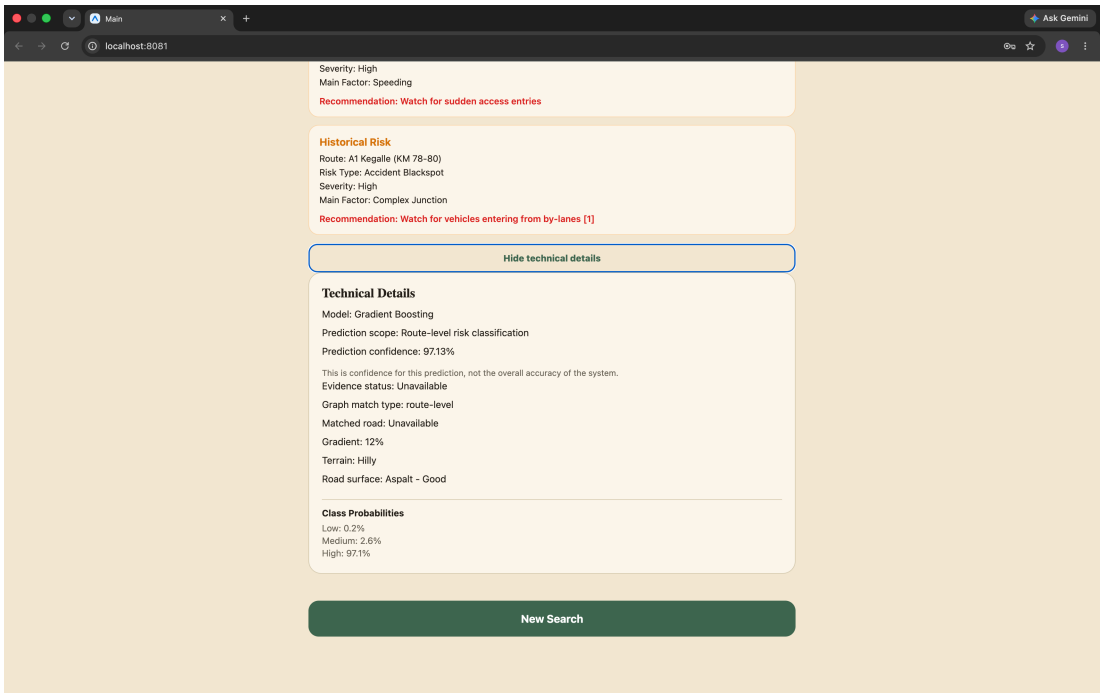


Figure 6.7. Gradient Boosting output and visible evidence-status fields.

Figure 6.7 displays prediction confidence of 97.13 percent and rounded class probabilities. This is confidence assigned to this prediction, not overall system accuracy. The panel also shows Evidence status and Matched road as Unavailable while displaying gradient 12 percent, Hilly terrain, and an Asphalt - Good surface. These fields require reconciliation before claiming complete evidence coverage.

6.3 User or stakeholder validation

No structured user study, expert-rating dataset, questionnaire results, or stakeholder interview analysis was supplied. Accordingly, the report does not claim measured usability, satisfaction, adoption, or expert agreement. A suitable next study would ask travelers and transport specialists to interpret the output, explain the recommendation, and identify uncertainty. Participant selection, consent, tasks, and scoring would need to be documented before reporting results.

6.4 Integration results

The supplied sequence shows route geometry, a severity output, current weather, vehicle suggestions, retrieved historical records, and technical details within the same local interface. The vehicle shortlist is consistent with the entered category and

passenger count, and the displayed prices fall within the stated budget. However, the screenshot sequence does not provide request identifiers or backend logs linking each capture. It should be presented as the author's demonstration sequence rather than an independently audited execution trace.

The technical panel shows Evidence status: Unavailable and Matched road: Unavailable alongside a 12 percent gradient and historical records. These visible status differences require verification of the relevant response fields. The screenshots are preserved as supplied; this report does not infer full evidence coverage from them.

6.5 Summary of findings

The stored classifier experiment supports internal severity-label prediction on the current grouped dataset. The interface demonstrates a Kandy-to-Colombo route and an affordable sedan shortlist with historical context. The example explicitly lacks alternative lower-risk comparison. Vehicle-rule transparency is described in the implementation, while the visible cards show a suitability label rather than the complete ranking trace. Field effectiveness and stakeholder validation remain unmeasured.

7 DISCUSSION

7.1 Interpretation of findings

The classifier reproduces severity-derived dataset labels. Its recorded holdout score is an internal result; it cannot verify the original observations or establish accident probability. Only 22.6 percent of training records have matching road profiles. The model may therefore depend on hazard context and missingness patterns, an interpretation requiring ablation and subgroup tests. Group separation reduces direct route-family overlap but does not resolve undocumented provenance, temporal changes, or limited geographic coverage.

7.2 Comparison with literature baselines and existing solutions

The implemented classifier follows the gradient-boosting family described by Friedman [5] and is compared with three alternative architectures on the project's own dataset. Gradient Boosting and Random Forest have close cross-validation means; only the predefined repeated-validation criterion supports the recorded selection. The report does not claim statistical superiority.

Routing work such as contraction hierarchies addresses efficient path computation [6], whereas this component adds historical severity and vehicle constraints to provider outputs. That distinction identifies the purpose of the integration but is not a benchmark result against routing systems. No comparative commercial-platform trial was conducted.

7.3 Research contribution

The research contribution is a reproducible prototype and an evaluation design separating route groups during model comparison and holdout testing. It examines a ten-feature severity model and preserves the distinction between measured road information, missing values, and contextual evidence. Its conclusions remain bounded by the quality and coverage of the source data. The work does not establish a new boosting algorithm or a causal model of road accidents.

7.4 Product contribution

The product contribution is a connected route-and-vehicle advisory workflow with explicit user constraints, historical context, recommendation outputs, and accessible technical details. The screenshots show how a traveler can move from an input form to a map and vehicle shortlist. The estimated Hyundai Verna trip price is below the example budget, and five seats meet the requested four-passenger capacity. The full suitability claim still depends on the recorded dataset attributes and implemented rules.

7.5 Limitations and threats to validity

Threats to validity include undocumented data provenance, severity-derived targets without external label validation, only 22 route groups, imbalanced classes, and missing road matches for 77.4 percent of training records. The road CSV lacks geometry, making road-family aggregation an approximation. Current weather is not a prediction of conditions during a future journey.

The screenshot demonstration has additional limits: alternative comparison is unavailable, no no-match or error-path screenshot is supplied, and technical evidence status needs reconciliation with other displayed values. No timing benchmark, user study, or prospective safety trial establishes operational effectiveness. These constraints should remain visible when the work is assessed.

7.6 Implications

Travelers and tourism operators could use the prototype for an initial vehicle shortlist and route-evidence summary. Commercial use would require verified inventory and prices, documented data rights, and service monitoring. No market-size, willingness-to-pay, accident-reduction, or user-satisfaction result is available. The demonstrated benefit is an integrated advisory interface; operational benefits remain to be evaluated.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research component integrates historical route hazard classification with vehicle recommendation for tourism planning in Sri Lanka. The prototype accepts journey constraints, retrieves available route context, predicts a severity class, and applies transparent rules to identify suitable vehicle candidates. Its explanation trace exposes the model inputs and recommendation rules, allowing results to be examined alongside missing evidence.

The recorded experiments select Gradient Boosting from four candidate classifiers through route-group cross-validation and repeated development validation. The selected model achieves a final holdout Macro F1 of 0.8780 and accuracy of 0.8820 on the current research dataset. These results support the feasibility of internal severity-label classification under the recorded evaluation design.

The study does not establish calibrated accident probability or nationwide road-safety accuracy. Unverified dataset provenance, sparse road-profile matches, and the lack of exact segment geometry restrict the strength of real-world conclusions. Vehicle ranking remains an attribute-based heuristic, and independent field evaluation has not demonstrated safety benefits.

8.2 Achievement of objectives

The prototype implements route evidence integration, a severity classifier, vehicle constraint rules, and explanatory outputs. Offline model comparison and reserved-holdout evaluation are recorded. The screenshots support the objective of presenting an integrated advisory flow for one example. The broader objective of demonstrating real-world recommendation quality is only partially addressed: independent data validation, expert assessment, and field evaluation remain future work.

8.3 Individual research and product contributions

The individual work brings route selection, vehicle selection, and recommendation explanations together within the tourism application. Its research aspect is the route-group classification workflow and interpretation of evidence limitations. Its product

aspect is the interface and backend integration that apply traveler constraints to the available road and vehicle data. Adjacent itinerary generation and accommodation functions remain separate team contributions.

8.4 Future directions

The first priority is to document and independently check dataset sources, permissions, collection methods, and labels. Road coverage should then be expanded using spatially referenced segments so that route analysis can be tied to the roads actually traversed.

Further experiments should report per-class confusion matrices, road-coverage subgroup performance, feature ablations, and temporal or geographic external validation. Probability calibration would require a suitable target and a separate evaluation design; it would not automatically convert severity-class outputs into accident probabilities.

Vehicle recommendation should be evaluated against expert-reviewed suitability cases and verified vehicle specifications. Field studies should record route conditions, recommendation feasibility, journey outcomes, and user interpretation with a defined protocol. Production improvements can include verified pricing, a persistent inference service, and measured latency and reliability. These are future tasks rather than completed results.

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GLOSSARY

Gradeability: The gradient capability recorded for a vehicle in the research dataset.

Gradeability margin: Vehicle gradeability minus the known road gradient.

Route family: A named or coded road grouping used to approximate route evidence.

Holdout: A partition reserved from model selection for final evaluation.

Macro F1: The unweighted mean of the class-specific F1 scores.

Predicted-class probability: The probability assigned by the classifier to its selected class; it is not calibrated accident probability.

APPENDICES

Appendix A Reproducibility notes

The stored experimental results are held in backend/src/ai-engine/data/model_comparison_results.csv and model_stability_summary.csv. The dataset cleaning report records class balance, route-group coverage, and missing road matches. These files provide the numerical basis for Chapter 6.

The model workflow is implemented in build_dataset.py, train_models_v2.py, and predict_safety.py under backend/src/ai-engine/scripts/. Route and vehicle integration is described in backend/SAFETY_ANALYZER_TECHNICAL_NOTES.md and implemented in the safety analysis service. Versioned copies of these artefacts should accompany future replication.

The demonstration inputs are Kandy to Colombo, LKR 25,000, four passengers, and Sedan. The original screenshots are presented only in Section 6.2. No screenshots are included in the appendices.

Reproduction requires the documented Python dependencies, configured backend and frontend services, and access to any enabled routing, geocoding, graph, and weather providers. External service outputs may change between runs. The reported model metrics are stored experiment outputs; this report preparation did not rerun model training.

Appendix B Academic integrity and AI use disclosure

Generative AI assistance was used through Codex to prepare and revise this report. The assistance included inspecting the project source and stored evaluation files, organising content against the supplied thesis template, drafting explanations, checking selected references, inserting the author-supplied screenshots, and rendering the document for layout review.

Representative requests were to prepare an individual report for IT22555458, follow the supplied template, and insert the screenshots without editing them. The affected artefacts are the report text, captions, tables, and document formatting. Screenshots were supplied by the author and embedded unchanged. Report preparation did not generate new model-training results, user-study outcomes, or field measurements.

The candidate should review every claim, citation, contribution statement, and institutional requirement before signing the declaration. This appendix records assistance visible in the present report-preparation workflow; any additional AI use during the research or software development must be added by the candidate.